

This is a project within a master course, which means that the student is given considerable freedom to construct his/her report. The grade of the report will therefore be given based on an overall assessment, rather than a list of “checked items”. Nevertheless, some report suggestions are provided below. Some parts, such as the addition of the code (appendix), are requirements. A “typical” report will be 5-10 pages, plus an appendix containing the (commented) code.

Report suggestions at the “approved” level (grade G)

- A model description (note that N must be at least 100).
- A brief description of the Metropolis Monte Carlo method.
- A description of how the simulation (plus code) was set up, and what sort of measures that were taken to speed up the simulation.
- A graph showing how the approach to equilibrium can be verified by plotting the energy versus the number of configurations. Do NOT plot the energy for ALL configurations – every 1/1000 or so will suffice (still a 1000 points, if you run an equilibration with a million configurations).
- Typical acceptance ratios, for attempted moves, should be quoted.
- Verified energy conservation (quote typical values for the relative energy drift – see Notes).
- Mention the (typical) number of configurations that were devoted to equilibration, as well as production runs.
- Equilibrium structures (2D-graphs), as well as radial distribution functions, for the various densities, at $T^*=0.1$
- A plot of how C_V (or just the energy fluctuations) vary with temperature, in the second part of the project (the gradual temperature drop), plus a description of this procedure.
- The code, with comments.

Suggested additions, to reach “VG” grade

- Successful runs of at least $2 \cdot 10^6$ configurations devoted to equilibration and $2 \cdot 10^7$ configurations devoted to production, for each new case (temperature or density) studied (N at least 100).
- Reasonable interpretations and discussions of the results obtained.
- A verified peak of the energy fluctuations, in the second part.
- Comparisons (mainly qualitative) with corresponding results in the Nature publication, including some comments.
- A discussion on model/method limitations, and sources of error